GATE ME 90 Days Challenge

Day 50/90 · Practice Set with Detailed Solutions

SUBJECT	TODAY'S TOPIC	COUNTDOWN
Applied Thermodynamics	**Refrigeration & Air Conditioning**	**53 days to GATE**

How to use this set: 15 questions (8 MCQ, 7 NAT) on vapour-compression and air (Bell–Coleman) refrigeration, heat pumps, COP and tonnes of refrigeration, vapour absorption, properties of moist air (specific and relative humidity, enthalpy, wet-bulb and dew point), the psychrometric chart and basic air-conditioning processes with bypass factor. Try them in **45 minutes** without looking at the solutions, then check the answer key and the step-by-step solutions. For NAT questions, enter the number; there is no negative marking for NAT in GATE.

Today's test: open 7 to 9 PM in the **GATE ME 90 Days Challenge** course on our app and website (enrol once for ₹1: iurlk.courses.store/909124), 20 new questions on the same topic. Top scorers are announced in the next 8 AM challenge reel.

Questions

Q1 MCQ

In a vapour-compression refrigeration cycle, subcooling the liquid refrigerant leaving the condenser (all other states unchanged)

(A) reduces the refrigerating effect

(B) increases the refrigerating effect and the COP without changing the compressor work per kg

(C) increases the compressor work per kg

(D) has no effect on the cycle

Q2 NAT

A reversed Carnot heat pump delivers heat to a room at 50 °C while taking heat from the ground at 5 °C. Its COP (round off to two decimal places) is _____

Q3 MCQ

One tonne of refrigeration (1 TR) is defined as the rate of heat removal needed to

(A) cool 1 tonne of water from 100 °C to 0 °C in 24 hours

(B) evaporate 1 tonne of water in 1 hour

(C) freeze 1 short ton (2000 lb) of water at 0 °C into ice at 0 °C in 24 hours

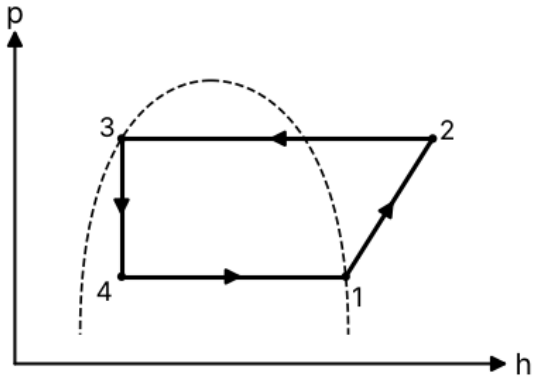
(D) freeze 1 tonne of water in 1 hour

Q4 NAT

A vapour-compression refrigerator works with these enthalpies: evaporator exit h_1 = 388 kJ/kg, compressor exit h_2 = 421 kJ/kg and condenser exit h_3 = 236 kJ/kg. The COP (round off to two decimal places) is _____

Q5 MCQ

The p–h diagram of a simple vapour-compression cycle is shown. The throttling (expansion) process is



(A) 3–4
(B) 1–2
(C) 2–3
(D) 4–1

Q6 NAT

An ideal Bell–Coleman air-refrigeration cycle works with a pressure ratio of 4 ($\gamma = 1.4$). Its COP (round off to two decimal places) is _____

Q7 MCQ

In an ammonia–water vapour-absorption refrigeration system, the compressor of the vapour-compression system is replaced by

(A) an expansion valve and a condenser
(B) a turbine and an evaporator
(C) a rectifier only
(D) an absorber, a solution pump and a generator

Q8 NAT

Air at 35 °C and a total pressure of 101.325 kPa has a relative humidity of 40 %. The saturation pressure of water at 35 °C is 5.628 kPa. The specific humidity of the air (in g per kg of dry air, round off to two decimal places) is _____

Q9 MCQ

For air–water vapour mixtures at ordinary conditions, the thermodynamic wet-bulb (adiabatic saturation) temperature is practically equal to the wet-bulb temperature read by a psychrometer because

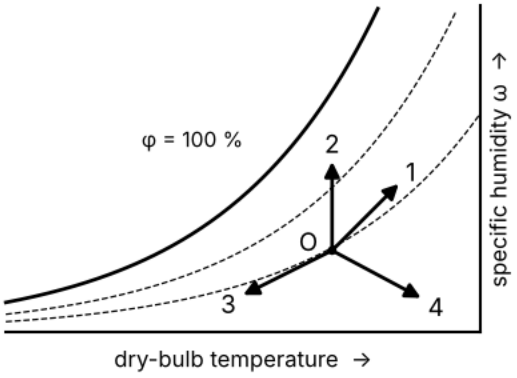
(A) the air is always saturated
(B) the Lewis number for air–water vapour is close to 1
(C) the specific heat of water is constant
(D) the dew point equals the dry-bulb temperature

Q10 NAT

Moist air at 30 °C has a specific humidity of 0.015 kg/kg of dry air. Taking $h = 1.005t + \omega(2500 + 1.88t)$ kJ/kg of dry air (t in °C), its enthalpy (in kJ/kg of dry air, round off to two decimal places) is _____

Q11 MCQ

The schematic psychrometric chart shows four processes from the state O. Chemical (adsorption) dehumidification, e.g. over silica gel, is best represented by



(A) O–1
(B) O–2
(C) O–3
(D) O–4

Q12 NAT

Air at 30 °C flows over a cooling coil whose apparatus dew point is 12 °C. If the bypass factor of the coil is 0.15, the temperature of the air leaving the coil (in °C, round off to one decimal place) is _____

Q13 MCQ

A vapour-absorption refrigerator receives heat in its generator at 100 °C, rejects heat to the surroundings at 30 °C and produces cooling at −10 °C. Its maximum possible COP is about

(A) 1.23
(B) 6.58
(C) 0.19
(D) 5.33

Q14 NAT

Moist air with 1.2 kg/s of dry air and a specific humidity of 0.006 kg/kg is heated from 10 °C to 30 °C in a heating coil. Taking c_p of dry air = 1.005 and of water vapour = 1.88 kJ/kg·K, the heating load (in kW, round off to two decimal places) is _____

Q15 MCQ

The refrigerant vapour leaving the evaporator of a practical refrigerator is usually slightly superheated mainly to

(A) raise the condenser pressure
(B) increase the refrigerating effect per kg by a large amount
(C) make sure that no liquid droplets enter the compressor
(D) reduce the compressor discharge temperature

Answer Key

Q	Type	Answer
Q1	MCQ	(B) increases the refrigerating effect and the COP without changing the compressor work per kg
Q2	NAT	7.18 (accept 7.14 to 7.22)
Q3	MCQ	(C) freeze 1 short ton (2000 lb) of water at 0 °C into ice at 0 °C in 24 hours
Q4	NAT	4.61 (accept 4.58 to 4.64)
Q5	MCQ	(A) 3–4
Q6	NAT	2.06 (accept 2.04 to 2.08)
Q7	MCQ	(D) an absorber, a solution pump and a generator
Q8	NAT	14.13 (accept 14.06 to 14.20)
Q9	MCQ	(B) the Lewis number for air–water vapour is close to 1
Q10	NAT	68.50 (accept 68.15 to 68.85)
Q11	MCQ	(D) O–4
Q12	NAT	14.7 (accept 14.6 to 14.8)
Q13	MCQ	(A) 1.23
Q14	NAT	24.39 (accept 24.27 to 24.51)
Q15	MCQ	(C) make sure that no liquid droplets enter the compressor

Detailed Solutions

Q1 Answer: (B) increases the refrigerating effect and the COP without changing the compressor work per kg

Concept: The throttle is isenthalpic, so $h_4 = h_3$.

Subcooling lowers h_3, hence h_4, so the refrigerating effect $h_1 - h_4$ grows.

Compressor inlet and outlet states are unchanged, so the work per kg is the same: COP rises **(B)**.

Exam tip: A liquid–suction heat exchanger gives subcooling, but the extra suction superheat it causes may raise the work.

Q2 Answer: 7.18 (accept 7.14 to 7.22)

Concept: $COP_{HP} = T_H/(T_H - T_L)$, with absolute temperatures.

$T_H = 323.15$ K, $T_L = 278.15$ K, $T_H - T_L = 45$ K

$COP_{HP} = 323.15/45 =$ **7.18**

Exam tip: The refrigerator COP between the same temperatures is 278.15/45 = 6.18, exactly one less.

Q3 Answer: (C) freeze 1 short ton (2000 lb) of water at 0 °C into ice at 0 °C in 24 hours

Concept: The unit comes from the ice industry.

Freezing 907 kg of water at 0 °C (latent heat ≈ 335 kJ/kg) in 24 h needs ≈ 3.5 kW: **(C)**.

1 TR = 211 kJ/min = 3.517 kW (often taken as 3.5 kW).

Exam tip: Always check which value of 1 TR the question gives (3.5 or 3.517 kW).

Q4 Answer: 4.61 (accept 4.58 to 4.64)

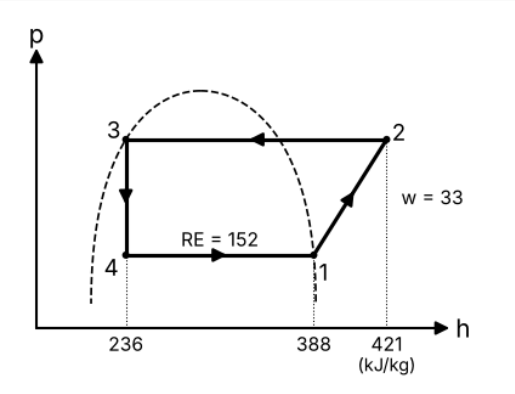
Concept: COP = refrigerating effect/compressor work = $(h_1 - h_4)/(h_2 - h_1)$, with $h_4 = h_3$.

Refrigerating effect = 388 − 236 = 152 kJ/kg

Compressor work = 421 − 388 = 33 kJ/kg

COP = 152/33 = **4.61**

Exam tip: The heat rejected in the condenser is $h_2 - h_3$ = 185 kJ/kg = 152 + 33, a quick check.



Q5 Answer: (A) 3–4

Concept: Throttling is isenthalpic, so it is a vertical line (h constant) from high to low pressure on the p–h chart.

3–4 drops from condenser pressure to evaporator pressure at constant h: **(A)**.

1–2 is compression, 2–3 condensation (heat rejection) and 4–1 evaporation (refrigerating effect).

Exam tip: On a T–s chart the throttle is drawn dashed, because it is irreversible.

Q6 Answer: 2.06 (accept 2.04 to 2.08)

Concept: COP = $1/(r_p^{(\gamma-1)/\gamma} - 1)$ for isentropic compression and expansion.

$4^{0.2857} = 1.4860$

COP = 1/(1.4860 − 1) = **2.06**

Exam tip: The COP depends only on the pressure ratio; reducing r_p raises COP but also cuts the refrigerating effect per kg of air.

Q7 Answer: (D) an absorber, a solution pump and a generator

Concept: The absorption system compresses the refrigerant 'thermally'.

Ammonia vapour is absorbed in weak solution in the absorber, the strong solution is pumped to high pressure, and heat in the generator drives the ammonia off: **(D)**.

Pumping a liquid needs very little work compared with compressing a vapour.

Exam tip: In a LiBr–water system, water is the refrigerant and LiBr the absorbent; in NH_3–water, ammonia is the refrigerant.

Q8 **Answer: 14.13 (accept 14.06 to 14.20)**

Concept: $p_v = \varphi p_s$; $\omega = 0.622\, p_v/(p - p_v)$.

$p_v = 0.40 \times 5.628 = 2.251$ kPa

$\omega = 0.622 \times 2.251/(101.325 - 2.251) = 0.01413$ kg/kg = **14.13 g/kg**

Exam tip: Degree of saturation $\mu = \omega/\omega_s$ is slightly less than φ (here $\mu \approx 0.39$).

Q9 **Answer: (B) the Lewis number for air–water vapour is close to 1**

Concept: The psychrometric WBT is set by a balance of heat and mass transfer at the wick; the adiabatic saturation temperature by an energy balance.

The two coincide when $h/(\rho c_p h_m) = Le^{2/3} \approx 1$, true for air–water: **(B)**.

That is why constant-WBT lines on the chart are (almost) constant-enthalpy lines.

Exam tip: For other gas–liquid pairs (e.g. air–alcohol) the two temperatures differ.

Q10 **Answer: 68.50 (accept 68.15 to 68.85)**

Concept: Enthalpy of moist air = dry-air sensible part + vapour part (latent at 0 °C + superheat).

Dry air: $1.005 \times 30 = 30.15$ kJ/kg

Vapour: $0.015 \times (2500 + 1.88 \times 30) = 0.015 \times 2556.4 = 38.35$ kJ/kg

$h = 30.15 + 38.35$ = **68.50 kJ/kg of dry air**

Exam tip: More than half of this enthalpy is latent: removing moisture is what loads a cooling coil.

Q11 **Answer: (D) O–4**

Concept: In adsorption the vapour gives up its latent heat (plus heat of adsorption) to the air stream.

So ω falls while the dry-bulb temperature rises, roughly along a constant-enthalpy line: down and to the right, **O–4 (D)**.

O–1 is heating with humidification, O–2 humidification at constant DBT and O–3 cooling with dehumidification.

Exam tip: Cooling-coil dehumidification (O–3) cools the air; chemical dehumidification heats it.

Q12 **Answer: 14.7 (accept 14.6 to 14.8)**

Concept: $BF = (t_2 - t_{ADP})/(t_1 - t_{ADP})$: the fraction of air that 'misses' the coil surface.

$t_2 = 12 + 0.15 \times (30 - 12) = 12 + 2.7$

t_2 = **14.7 °C**

Exam tip: More coil rows or lower air velocity give a smaller bypass factor (higher contact factor $1 - BF$).

Q13 **Answer: (A) 1.23**

Concept: Ideal absorption system = reversible heat engine ($T_G \rightarrow T_0$) driving a reversible refrigerator ($T_E \rightarrow T_0$).

$\eta_{engine} = 1 - 303.15/373.15 = 0.1876$

$COP_{refrigerator} = 263.15/(303.15 - 263.15) = 6.579$

$COP_{max} = 0.1876 \times 6.579 =$ **1.23 (A)**

Exam tip: Real absorption COPs are about 0.5–0.7: they pay off only with cheap waste heat.

Q14 **Answer: 24.39 (accept 24.27 to 24.51)**

Concept: Sensible heating at constant ω uses the humid specific heat $c_{pa} + \omega c_{pv}$.

Humid specific heat = $1.005 + 0.006 \times 1.88 = 1.01628$ kJ/kg·K

$Q = 1.2 \times 1.01628 \times (30 - 10) =$ **24.39 kW**

Exam tip: Ignoring the vapour term gives 24.12 kW, which is outside the accepted range here.

Q15 **Answer: (C) make sure that no liquid droplets enter the compressor**

Concept: Liquid entering a reciprocating compressor (slugging) can damage valves and dilute the lubricant.

A few degrees of superheat, controlled by the thermostatic expansion valve, guarantees 'dry' compression: **(C)**.

Superheat raises the suction specific volume and the discharge temperature, so it is kept small.

Exam tip: Useful superheat (inside the cold space) adds to the refrigerating effect; superheat in the suction line does not.

What next?

Day 50 test tonight, 7 to 9 PM

20 questions · 60 minutes · GATE marking (1/3 and 2/3 negative). Top 3 scorers are announced with name, marks and photo (if shared) in the next 8 AM challenge reel. Not enrolled yet? Join the ₹1 challenge course: iurlk.courses.store/909124

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Next, Day 51/90: Engineering Mathematics, Complex Variables & Numerical Methods. Comment **GATE90** on our reel to get all links.
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